In the absence of external forces,

$$\mathbf{0} = \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{l} = \dot{\boldsymbol{\Theta}}\boldsymbol{\pi} + \boldsymbol{\Theta}\dot{\boldsymbol{\pi}}.$$

Therefore

$$\dot{\boldsymbol{\pi}} = -\boldsymbol{\Theta}^T\dot{\boldsymbol{\Theta}}\boldsymbol{\pi}.$$

Next observe that

$$\boldsymbol{\Theta}^T\dot{\boldsymbol{\Theta}} = \mathrm{skew}(\boldsymbol{\omega}_\mathrm{b}).$$

This follows from the definition of the angular velocity and properties of the cross product (see exercises). Thus

$$\dot{\boldsymbol{\pi}} = -\boldsymbol{\omega}_\mathrm{b} \times \boldsymbol{\pi} = \boldsymbol{\pi} \times \boldsymbol{\omega}_\mathrm{b},$$

but since $\boldsymbol{\omega}_\mathrm{b} = \boldsymbol{T}^{-1}\boldsymbol{\pi}$, we may conclude

$$\dot{\boldsymbol{\pi}} = \boldsymbol{\pi} \times \boldsymbol{T}^{-1}\boldsymbol{\pi}. \tag{4.41}$$

These are the so-called *Euler equations* for the free rigid body.

Let us observe that (4.41) is in the form

$$\dot{\boldsymbol{\pi}} = \tilde{\boldsymbol{J}}(\boldsymbol{\pi})\nabla\tilde{H},$$

where $\tilde{\boldsymbol{J}}(\boldsymbol{\pi}) = \mathrm{skew}(\boldsymbol{\pi})$ is obviously skew symmetric, and $\tilde{H} = \boldsymbol{\pi}^T\boldsymbol{T}^{-1}\boldsymbol{\pi}/2$ is the energy function. Thus the Euler equations are in the form of a Hamiltonian system on a three-dimensional phase space. The symplectic structure matrix is dependent on the phase variables, and it is degenerate: $\tilde{\boldsymbol{J}}\boldsymbol{\pi} = \boldsymbol{\pi} \times \boldsymbol{\pi} = \mathbf{0}$, signifying that total angular momentum is conserved: $\|\boldsymbol{\pi}\| = \text{constant}$. This is a generalization of our previous concept of *Hamiltonian system*. Much of the previously developed theory of Hamiltonian systems can be adapted to the setting of the free rigid body with analogous definitions of splitting method and symplectic integrators obtained by additive decomposition of the Hamiltonian.

It is possible to solve the Euler equations using an exact solution and to use this to design efficient integrators [68, 249, 382], but in the most common setting of molecular dynamics, where additional approximations will be need, we are usually justified in treating these equations using a more elementary procedure [258]. If $\boldsymbol{T}$ is a diagonal matrix, $\boldsymbol{T} = \mathrm{diag}(I_1, I_2, I_3)$, then we split the Euler equations and approximate the solution by successive rotations around the three axes of inertia,